

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVC603 - TRANSPORTATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 According to IRC classification, which category of road is primarily meant to connect district headquarters to each other?

- a) National Highways
- b) State Highways
- c) Major District Roads
- d) Village Roads

Which one of the following factors is most critical in deciding the alignment of a highway in hilly terrain?

- a) Population density
- b) Soil fertility
- c) Gradient and topography
- d) Availability of road construction materials

3 The super elevation 'e' required on a horizontal curve is given by which formula (neglecting friction)?

- a) $e = \frac{v^2}{gR}$
- b) $e = \frac{gR}{v^2}$
- c) $e = \frac{W \cdot R}{v^2}$
- d) $e = \frac{vR}{g}$

Which of the following cross-sectional elements is most critical for vehicle safety at high speeds?

- a) Camber
- b) Shoulder width
- c) Superelevation
- d) Kerbs

- 5 **Which of the following tests best determines the durability of aggregate under alternate wetting and drying conditions?**
a) Impact test
b) Soundness test
c) Crushing test
d) Flakiness Index test
- 6 **Which material has the highest softening point and is more suitable for hot climate pavements?**
a) Cutback bitumen
b) Tar
c) Bitumen of grade VG-40
d) Bitumen emulsion
- 7 **Which traffic volume study method is most appropriate for planning long-term public transport facilities?**
a) Spot speed studies
b) Manual vehicle count
c) Origin-destination survey
d) Parking survey
- 8 **In rotary intersections, which rule is applied to regulate traffic flow?**
a) Left lane priority
b) First come first serve
c) Priority to traffic inside the rotary
d) Priority to right turning traffic
- 9 **Anti-crash barriers are typically designed to:**
a) Increase speed of vehicles
b) Prevent waterlogging
c) Absorb kinetic energy and deflect vehicles back to the roadway
d) Improve drainage
- 10 **Which of the following is not a function of Intelligent Transportation Systems (ITS)?**
a) Real-time traffic monitoring
b) Dynamic signal control
c) Increase toll rates automatically
d) Emergency vehicle management
- 11 **In IRC: 58-2011 for rigid pavement design, the load transfer efficiency of dowel bars is assumed to be:**
a) 100%
b) 75%
c) 50%
d) 25%
- 12 **In flexible pavement design using IRC:37-2012, which factor is used to account for vehicle loading over time?**
a) Vehicle Damage Factor (VDF)
b) Load Distribution Factor (LDF)
c) Equivalent Axle Load Factor (EALF)
d) Contact Pressure Ratio (CPR)

Q2. Solve the following.

- A) What were the Jaykar Committee recommendations with its effect? also discuss the first twenty year road development plan in details. 6
- B) Write an explanatory note on Highway Planning Survey's. 6

- Q3. Solve the following.
- A) The speed of overtaken and overtaking vehicle is 80 and 60 kmph respectively on a one way traffic road. If acceleration of overtaking vehicle is 0.99 m/sec^2 . Calculate : i. Safe overtaking sight distance ii. Minimum length of Overtaking zone 6
- B) Discuss the construction and maintenance of highway drainage systems. 6
- Q4. Solve Any Two of the following.
- A) What are the tests performed on Bitumen? Explain any one in detail with procedure? 6
- Explain CBR test with sketch also state its advantages & limitation. 6
- Explain the relevant properties of soil used in highway construction. 6
- Q5. Solve Any Two of the following.
- A) What is Traffic Engineering? Explain traffic characteristics. 6
- B) Explain in detail i. Spot speed Studies ii. Origin and Destination Study 6
- C) Explain the methods used for the analysis and interpretation of traffic survey data 6
- Q6. Solve Any Two of the following.
- A) Give the comparison between rigid and flexible pavement in detail? 6
- B) Explain different components of pavement with its function? 6
- Provide an overview of other modes of transportation such as railways, airways, waterways, and pipeline transportation. 6
